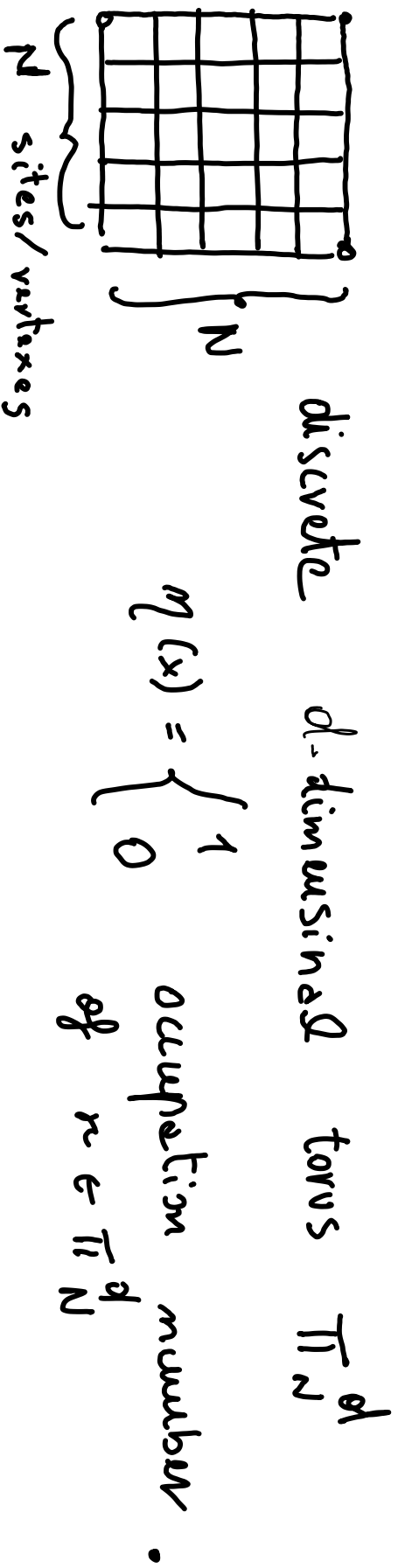


HYDRODYNAMICS OF DIFFUSIVE IPS

Lecture 1: Ch. 4 KL, Hydrodynamics of symmetric $\textcircled{1}$ simple exclusion process (SEP).



configuration $\underline{n} := \{n(x)\}_x \in \Sigma_N = \{0, 1\}^{\pi_N^d}$ state space.

occupied $\bullet$ empty $\circ$

a particle can jump from x to y if x is occupied and y is empty.

can not if both are occupied.

Contact S: 0038-340-5345065, LeoSch.st@protonmail.com

the stochastic dynamics according to a CTMC, defined by the Markov generator $\mathcal{L}_N$: (2)

$$(\mathcal{L}_N f)(n) = \sum_{(n,y) \in E_N} \underbrace{\eta(n)(1-\eta(y))}_{\text{exclusion rule}} [f(\eta^{n,y}) - f(n)]$$

$f: \Sigma_N \rightarrow \mathbb{R}$, E_N : set of oriented bonds/edges

$$\eta^{x,y}(z) := \begin{cases} \eta(x)-1, & z=x \\ \eta(y)+1, & z=y \\ \eta(z), & z \neq x,y \end{cases}$$

new configuration
after a jump on (x,y) .

Fact: $\mathcal{L}_N$ is Bernoulli reversible $\mu_N^\theta(\eta) = \prod_x \eta(x)^{\theta} (1-\eta(x))^{(1-\theta)}$ constant prob. l.e.

We want to derive the macroscopic behaviour of observables.

$$\underline{m}_t := \{m_t(n)\}_n \text{ configuration at time } t.$$

Def. (empirical measure (represents density of particles))

(3)

$$\pi_t^N(du, n) := \frac{1}{N^d} \sum_{n \in \pi_N^d} \overline{\eta_t(n)} \delta_{n_N}(du)$$

$\mathcal{H}_+(\pi^d)$ space of positive measures on $\overline{\pi}^d := [0, 1]^d$

Remark

two space scales:

- the microscopic one: π_n^d (edges of size 1)
- the macroscopic one: $\overline{\pi}^d$

$$x \in \pi_N^d \mapsto n_N \in \pi^d, \quad u \in \overline{\pi}^d \mapsto [Nu] \in \overline{\pi}_N^d$$

$$\int_0^1 \pi_t^N(du, n) f(u) = \frac{1}{N^d} \sum_{x \in \pi_N^d} f(n) \eta_t^+(n)$$

• Call $\mathcal{D}([0,1], \Sigma_N)$ space of ^{CADLAC} trajectories of the process $\{m_t\}_t$ and $\mathcal{Z}([0,1], \mathcal{H}_+(\pi^d))$ that of $\pi_t(m, du)$. (4)

• $\mu_{t=0}^{(n)}$ are distributed along an initial distribution μ_N .

• Remark We consider the accelerated process $\mu_{t(N^2)^{\leftarrow}}$ diffusive time scale or $N^2 \mathcal{L}_N$.
 $\hookrightarrow$ Two time scale: $\begin{cases} \text{macroscopic } [0, t] \\ \text{microscopic } [0, N^2 t] \end{cases}$

Def Let $\{P_N\}_N$ and $\{Q_N\}_N$ respectively the sequences of probability measures induced by $\{\mu_{N^2 t}\}_t$ on $\mathcal{D}([0,1], \Sigma_N)$ and by $\pi_t^N(du, m)$ on $\mathcal{Z}([0,1], \mathcal{H}_+(\pi^d))$.

Statement of Hydrodynamics

(5)

$\rho_0 : \mathbb{T}^d \rightarrow \mathbb{R}^+, 1$ initial ρ profile (smooth)
 $\{\mu_N^t\}$ the associated sequence of Bernoulli
 initial measures, i.e. $\mu_N^t(\eta) = \prod_{x \in \mathbb{T}_N^d} [\eta(x)]^{\rho_0(x/N)} [1 - \eta(x)]^{(1 - \rho_0(x/N))}$.

Then $\forall t \geq 0$, the sequence of random
 measures $\pi_t^N(du, n)$ converges in Probability
 to the absolute continuous measures $\rho(t, u) du$, i.e. .

$$\lim_{N \rightarrow \infty} \mathbb{P}_N \left(\left| \int_{\mathbb{T}^d} f d\pi_t^N(\eta) - \int_{\mathbb{T}^d} f \rho(u, t) du \right| > \varepsilon \right) = 0,$$

$\forall \varepsilon > 0, f \in C(\mathbb{T}^d)$, where $\rho(t, u)$ is the solution (strong) of
 the Cauchy problem $\begin{cases} \partial_t \rho = \Delta \rho \\ \rho(0) = \rho_0 \end{cases}$.

$$\leadsto \left[\frac{1}{N} \int_{\mathbb{T}^d} f(\eta, du) \xrightarrow{(N)} \rho(u, t) \right] \text{ as } \underline{L-L-N}$$

Steps of the proof

⑥

- 1) Proving that $\{Q_N\}_N$ is relatively compact [Functional Probability]
- 2) Identifying the limit points and proving that all converging subsequences $\{Q_{N_k}\}_k$ have the same limit.

Remarks

- A) Point 1) requires to set a topology on $\mathcal{C}([0, T], \mathcal{H}^1)$ to characterize compact set
- B) Point 2) will be given by the typical behaviour [in probability (LLN)] of $\pi_t^N(\eta, du)$ for $N \gg 1$, i.e. the PDE characterizing the hydrodynamics.

Topology and Compactness

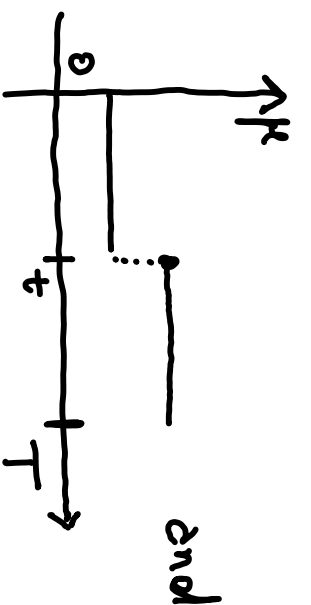
$\mathcal{H}^+(\pi^d)$ is sep. complt metric space with the metric induced by the weak top.

$$g(\mu, \nu) := \sum_{k=1}^{\infty} \frac{1}{2^k} \frac{|\langle \mu, f_k \rangle - \langle \nu, f_k \rangle|}{1 + |\langle \mu, f_k \rangle - \langle \nu, f_k \rangle|}, \quad \{f_k\}_k \text{ dense countable of } C(\pi^d)$$

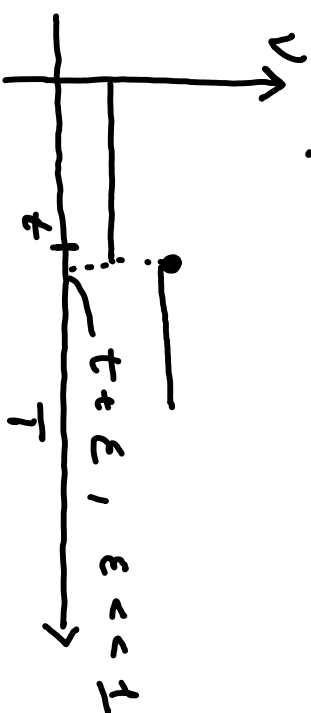
$$\langle f, \mu \rangle := \int \pi^d f \, d\mu.$$

Remark On $\mathcal{C}([0, T], \mathcal{H}^+)$ the uniform topology
 $d(\mu, \nu) = \sup_{s, t \in [0, T]} g(\mu(s), \nu(t))$ is not good because

it evaluates as "very different" measures that differ for a small time difference in a jump, like



and



Billingsley, "Convergence of Probability Measures", ch 3, §12

(4)

$\leadsto$ So it is introduced the Skorohod Topology: 8
 two trajectories $\mu(s)$ and $\nu(s)$ are "close" if deforming
 a little bit of the same time their "ordinate/graph"
 and "time scale" they are "similar".

$$d(\mu, \nu) := \inf_{\lambda \in \Lambda} \max \left\{ \sup_{0 \leq t \leq T} |\lambda(t) - t|, \sup_{0 \leq t \leq T} g(\mu(t), \nu(\lambda(t))) \right\}$$

where $\Lambda := \left\{ \begin{array}{l} \text{strictly increasing functions } \lambda: [0, T] \rightarrow [0, T] \\ \text{with } \lambda(0) = 0 \text{ and } \lambda(T) = T \end{array} \right\}$

$\leadsto \mathcal{C}$ becomes sep. complete 8 metric space.

~~8~~ for real for completeness something slightly different has
 to be considered,

Functional Ascoli-Arzelà: Given $A \subset \mathcal{C}([0, T], \mathcal{H}^+)$ if

1) $\{ \mu_t ; \mu \in A, t \in [0, T] \}$ is relatively compact on $\mathcal{H}^+$ ⑧

[boundedness property, $\forall \subset \mathcal{H}^+$ is rel. comp. iff $\sup_{\mu \in \mathcal{H}} \int_{\mathbb{R}^d} d\mu(x) < +\infty$, i.e. total mass is finite.]
 (total variation norm $\|\mu\|$ of μ .)

2) $\lim_{\delta \rightarrow 0} \sup_{\mu \in A} W_\mu(\delta) = 0$, $W_\mu(\delta) := \sup_{|t-s| \leq \delta} g(\mu_s, \mu_t)$
 [modulus of continuity]

[uniform equicontinuity property, μ is uniformly continuous iff $\lim_{\delta \rightarrow 0} W_\mu(\delta) = 0$.]

$\Rightarrow A$ is relatively compact.

Remark: We have to reduce this in probabilistic terms $\leadsto$ Prokhorov's theorem for $\{\mathcal{P}_N\}_N$.

Reminder (Prohorov's theorem):

Let X compl. sep. metric space and $\{\mu_N\}_N$ seq. of prob. on it. (10)

$\{\mu_N\}_N$ is tight iff $\{\mu_N\}_N$ relatively compact

$$[\forall \varepsilon > 0 \exists K_\varepsilon \mid \mu_N(K_\varepsilon) \geq 1 - \varepsilon \quad \forall N] \quad [\exists \{\mu_{N_k}\}_{N_k} \mid \mu_{N_k} \xrightarrow{w} \mu]$$

Functional Prohorov (to prove point 1) of Hydro.)

Let $\{\mu_N\}_N$ sequence of pr. meas. on $\Sigma([0, T], \mathcal{H}^+)$ if

1) $\forall t \in [0, T]$ and $\forall \varepsilon > 0$, there is a compact

$$K(t, \varepsilon) \subset \mathcal{H}^+ \text{ s.t. } \sup_N \mu_N[\mu: \mu \notin K(t, \varepsilon)] \leq \varepsilon$$

(i.e. we have a tightness of $\{\mu_N\}_N$)

2) $\forall \varepsilon > 0$, $\lim_{\delta \rightarrow 0} \limsup_{N \rightarrow \infty} \mu_N[\mu: \mu(\delta) > \varepsilon] = 0$,
(i.e. equicontinuity in probability) μ (all trajectory $\mu \in \mathcal{H}^+$)

$\Rightarrow \{\mu_N\}_N$ is relatively compact.

Concrete meaning: what to do

(11)

For 1) we show boundedness of the total variation norm on all time

$$\text{i.e.} \quad \lim_{\ell \rightarrow \infty} \limsup_{N \rightarrow \infty} \mathcal{V}_N \left\{ \sup_{0 \leq t \leq T} \frac{1}{N} \sum_x |\eta_t(x)| \geq \ell \right\} = 0$$

[deterministic for SEP]

For 2) we show uniform equicontinuity for each test function $f \in C(\pi^N)$

$$\text{i.e.} \quad \forall \varepsilon > 0 \quad \lim_{\delta \downarrow 0} \limsup_{N \rightarrow \infty} \mathcal{P}_N \left\{ \sup_{\substack{0 \leq s, t \leq T \\ |t-s| \leq \delta}} | \langle f, \pi_t^N \rangle - \langle f, \pi_s^N \rangle | \geq \varepsilon \right\} = 0$$

$\{f_k\}$ countable dense set of $C(\pi^d)$.

1) is granted for SSEP

(12)

2) Consider $C^2(\mathbb{R}^d)$ dense set and the following martingale

$$t \geq s \geq 0, M(t-s) = \langle \pi_t^N, f \rangle - \langle \pi_s^N, f \rangle - N^2 \int_s^t dr \mathcal{L}_N \langle \pi_r^N, f \rangle \left[\begin{array}{l} \text{Dynkin's} \\ \text{formula} \end{array} \right]$$

$$Q_N \left(\sup_{\substack{0 \leq s, t \leq T \\ |t-s| < \delta}} |\langle \pi_t^N, f \rangle - \langle \pi_s^N, f \rangle| > \varepsilon \right)$$

from Dynkin

$$\leq Q_N \left(\sup_{\substack{0 \leq s, t \leq T \\ |t-s| < \delta}} |M(t,s)| + N^2 \left| \int_s^t \mathcal{L}_N \langle \pi_r^N, f \rangle \right| > \varepsilon \right)$$

$$\begin{aligned} &= \underbrace{\frac{1}{2} \int_s^t dr \langle \pi_r^N, \Delta_N f \rangle}_{\text{discrete Laplacian of } f} + \underbrace{N^2 \left| \int_s^t \mathcal{L}_N \langle \pi_r^N, f \rangle \right|}_{\text{constant}} \\ &\leq \underbrace{C(f)}_{\text{constant}} |t-s| \end{aligned}$$

For a CTHC given the martingale

$$M(t) = F(\eta_t) - F(\eta_0) - \int_0^t dr \, dF(\eta_r)$$

$$M(t) = M^2(t) - \underbrace{\int_0^t dr \left(\int_0^r F^2(\eta_v) - 2F(\eta_v) dF(\eta_v) \right)}_{L := B(s)}$$

is also a martingale.

Important trick:

$$(LF^2 - 2FLF)(\eta) = \frac{1}{2} \langle \eta, c(\eta, \eta') (F_{\eta'} - F_{\eta})^2 \rangle$$

$$\text{take } F(\eta_r) = \langle \pi_r'(\eta), f \rangle$$

$$E_{\eta_n}(M^2(t-s)) = E_{\eta_n} \left(\int_s^t B(s) ds \right) \leq c(f) \frac{|t-s|}{N_d}$$

$M(t)$ is martingale with initial value zero

$$\mathbb{P}_N \left(\sup_{\substack{0 \leq t, s \leq T \\ |t-s| > \delta}} |H(t-s)| + \left| N^2 \int_s^t \mathcal{L}_N < \tilde{\pi}_r^N(m), f > | \geq \varepsilon \right) \quad (14)$$

$$\leq \mathbb{P}_N \left(\sup_{\substack{0 \leq t, s \leq T \\ |t-s| > \delta}} |H(t-s)| > \frac{\varepsilon}{2} \right) + \mathbb{P}_N \left(\sup_{\substack{0 \leq t, s \leq T \\ |t-s| > \delta}} \left| N^2 \int_s^t \mathcal{L}_N < \tilde{\pi}_r^N(m), f > | > \frac{\varepsilon}{2} \right) \right.$$

from previous estimates and Chebyshev + Doob's ineq.

$$\text{We have that } \lim_{\delta \uparrow 0} \limsup_{N \rightarrow \infty} \mathbb{P} \left(\sup_{\substack{0 \leq t, s \leq T \\ |t-s| > \delta}} | \right. + \left. | > \varepsilon \right) = 0.$$

Second Step: characterization of limit points (hydrodynamics) (15)

Remark: $\sup_{0 \leq t \leq T} |\langle \pi_t^N, f \rangle| \leq \frac{1}{N^d} \sum_x f(x/N) \approx \int_{\mathbb{R}^d} f(u) du + O\left(\frac{1}{N}\right)$

$\mathcal{Q}_{Nk} \xrightarrow{(w)} \mathcal{Q}_k^* \leftarrow \text{limit point, } \mathcal{Q}_k^* [\pi: \pi_t(du) = \pi_t(u)du] = 1$
 $\xrightarrow{\text{concentrated on obs. c. traj. w.r.t. to Lebesgue.}}$

Consider the martingale

$$\tilde{M}(t) = \langle \pi_t^N, f \rangle - \langle \pi_0^N, f \rangle - \int_0^t ds \left[\langle \pi_s^N, \partial_s f \rangle + \mathcal{L}_N \langle \pi_s^N, f \rangle \right]$$

Define $\tilde{M}(t)$ and $\tilde{B}(t)$ as before we have

$$\lim_{N \rightarrow \infty} \sup \mathbb{P}_N(|\tilde{M}(t)| > \varepsilon) = 0$$

Remind

$$N^2 \mathcal{L}_N \langle \pi_s^N, f \rangle = \frac{N^2}{2} \langle \Delta_N f, \pi_s^N \rangle$$

$\hookrightarrow$ non trivial hydrodynamics

$$\frac{N^2}{2} \Delta_N f(x) \approx \frac{\Delta}{2} f(x) + O\left(\frac{1}{N^2}\right)$$

- Therefore the RHS is a discrete form of weak heat eq.:

(16)

$$M(t) = \frac{1}{N^{\alpha}} \sum_x m_t^n(x) f(x) - \frac{1}{N^{\beta}} \sum_x \eta_t^n(x) g(x) - \int_0^t ds \left[\bar{m}_s(x) \partial_s f(x) + \frac{N^2}{2} \sum_x \Delta f^n(x) \eta_s^n(x) \right]$$

$N \rightarrow \infty$ becomes a weak heat equation, with the initial condition

$$\int_{\mathbb{T}^d} \rho(0, x) f(x) = \int_{\mathbb{T}^d} du \rho_0(x) f(x)$$

because from the initial distribution $M(\eta) = \bar{1} \bar{1} (\eta(x))^{p_0(x)} (1 - \eta(x))^{(1-p(x))}$,

$$\lim_{N \rightarrow \infty} \mu_N \left(\left| \int_{\mathbb{T}^d} f d\pi_0^N(\eta) - \int_{\mathbb{T}^d} f \rho_0(x) dx \right| > \varepsilon \right) = 0, \forall \varepsilon > 0.$$

- So the limit points $Q^{\mathbb{R}}$ are concentrated on sol. of heat eq., moreover since the sol. is unique (and strong) $\leadsto$ so $Q^{\mathbb{R}}$ is unique, i.e.

$$\underbrace{Q_N}_{\text{subsequence}} \xrightarrow[N_K]{(N)} \int \rho(\varepsilon, u) du \sim \text{delta measure on the unique sol. of } \begin{cases} \partial_t \rho = \frac{1}{2} \partial_{xx} \rho \\ \rho(0, u) = \rho_0(u) \end{cases}$$

- But weak convergence to a deterministic value implies convergence in probability, i.e. $\pi_n^t(n, da) \xrightarrow{P} \delta(t, a) da$.

(17)

Reminder

Prop. Given a sequence $(X_n, \Omega_n, \mathcal{F}_n, \mathbb{P}_n)$ with (S, d) metric space and $X: \Omega_n \rightarrow S$, s.t. $X_n \xrightarrow{(w)} a \in S$, where $a \in S$ is a deterministic value, then $\mathbb{P}_n(d(X_n, a) < \varepsilon) \xrightarrow{P} 1$.
Namely $X_n \xrightarrow{(w)} a$ implies $X_n \xrightarrow{P} a$.

Proof. take $f = \chi_G$ where χ_G is the characteristic function of the open set $G = \{u \in S : d(u, a) < \varepsilon\}$,

$$\mathbb{E}_n(\chi_G(X_n)) \xrightarrow{P} \mathbb{E}(\chi_G(a)) = 1$$

$$\int_{\Omega_n} \chi_G(X_n) d\mathbb{P}_n = \mathbb{P}_n(X_n \in G)$$

□

OBSERVATION: Write the discrete heat eq.

(18)

$$M(t) = \frac{1}{N_0!} \sum_x \eta_x^1(x) f(x) - \frac{1}{N^d} \sum_x \eta_0(x) f(x) - \int_0^t ds \left[\sum_N \left[\sum_x \eta_s^1(x) \partial_s f(x) - N^2 \sum_x \eta_s^1(x) \Delta u f(x) \right] \right]$$

$\langle \pi_{t,1}^N, f \rangle$ $\langle \pi_0^N, f \rangle$ $\langle \pi_s^N, \partial_s f \rangle$ $\langle \pi_s^N, \Delta f \rangle$

• It is closed in terms of $\pi_t^N(\eta, du)$, in general it is not true. You can have $\eta_s^2(x)$ in place of $\eta_s(x)$

• In a more general case we proved we can replace $\eta_s(x)$ with $E_{P_{int}}(\eta_s(x))$ where the expectation is w.r.t. to Bernoulli of constant parameter $c(n, t)$.

So when we will have $\eta_s^2(x)$ we will replace this with $E_{P_{int}, t}(\eta_s^2(x)) = c^2(x, t)$, assuming the equilibrium is still a Bernoulli(c).

[the stationary measure is Bernoulli constant]